



**AKDENIZ UNIVERSITY**  
Faculty of Engineering · Department of Computer Engineering  
**CSE 427 — Wireless Sensor Networks**  
Midterm Examination (Practice) · Fall 2026-2027



**PRACTICE PAPER — prepared from the syllabus, the week 1 slides and the course textbooks (Karl & Willig; Dargie & Poellabauer). Not issued by the instructor. Scope: weeks 1-7.**

|                     |                   |                    |            |
|---------------------|-------------------|--------------------|------------|
| <b>Name Surname</b> |                   | <b>Student No.</b> |            |
| <b>Date</b>         | ___ / ___ / 20___ | <b>Signature</b>   |            |
| <b>Duration</b>     | 90 minutes        | <b>Total</b>       | 100 points |

**Instructions**

1. Closed book. A non-programmable calculator is allowed. Use  $\log_{10}$  where logarithms appear.
2. Show all work. A correct answer without justification receives no credit.
3. Answer in the space provided. If you need more space, use the back of the page and say so.

| Q            | 1  | 2  | 3  | 4  | 5  | 6  | Total |
|--------------|----|----|----|----|----|----|-------|
| <b>Max</b>   | 14 | 14 | 18 | 20 | 20 | 14 | 100   |
| <b>Score</b> |    |    |    |    |    |    |       |

**Question 1 — WSN Fundamentals and Application Taxonomy**

*14 points*

- (a) Define a *mote* and a *sink / gateway*. Give two reasons why a WSN is not "a normal Wi-Fi network made smaller".

*4 pts*

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A greenhouse WSN produces the four message types below.

| #  | Message                                             |
|----|-----------------------------------------------------|
| M1 | Temperature and humidity every minute               |
| M2 | "Overheating!" alarm the moment 40 °C is exceeded   |
| M3 | Operator asks a node for its battery level          |
| M4 | Gateway commands an actuator node to open the vents |

- (b) Classify each message as periodic, event-driven, query-driven or control. For each, state whether it most needs **low latency**, **high reliability**, **authentication**, or none of these, with one reason. 6 pts

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- (c) Why do security and safety "stop being extra topics" once a WSN controls actuators? Give one concrete attack on M4. 4 pts

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**Question 2 — Mote Hardware and Contiki** 14 points

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- (a) Draw the block diagram of a mote with the seven blocks from the lecture and the direction of data flow. Mark the two blocks that dominate node lifetime. 5 pts

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Consider the Contiki 3.0 process below, compiled for a Sky mote.

```
PROCESS(sense_process, "Sense");
AUTOSTART_PROCESSES(&sense_process);
PROCESS_THREAD(sense_process, ev, data)
{
    static struct etimer t;
    PROCESS_BEGIN();
    while(1) {
        etimer_set(&t, CLOCK_SECOND * 5);
        PROCESS_WAIT_EVENT_UNTIL(etimer_expired(&t));
        printf("sample\n");
    }
    PROCESS_END();
}
```

- (b) How often is "sample" printed, and in which Cooja window do you see it? Why must **t** be declared **static**? Why is this better than a busy-wait loop on a mote? 5 pts

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- (c) In Cooja, compare a **Sky mote** with a **Cooja mote**: what does each simulate, and when would you choose each? 4 pts

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### Question 3 — Radio and Physical Layer

18 points

A mote transmits at  $P_t = 0$  dBm. Use the log-distance path-loss model with reference loss  $PL(d_0 = 1 \text{ m}) = 40$  dB and path-loss exponent  $n = 3$ :

$$PL(d) = PL(d_0) + 10 \cdot n \cdot \log_{10}(d / d_0) + X_\sigma$$

The receiver sensitivity is  $-94$  dBm. Ignore  $X_\sigma$  in (a) and (b).

- (a) Compute the received power at 20 m. 4 pts

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- (b) Compute the maximum communication range. 5 pts

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- (c) Now add log-normal shadowing,  $X_\sigma \sim N(0, 4^2)$  dB. What is the probability that a link at the range from (b) works? At what distance does a link succeed with probability 0.90? (Use  $z_{0.90} = 1.28$ .) 5 pts

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- (d) Define RSSI, LQI and PRR. Explain the *transitional region* and why links can be asymmetric. 4 pts

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#### Question 4 — Energy

20 points

A Sky-class mote runs on two AA cells with a usable capacity of **2500 mAh**. The node is always in exactly one of the states below.

| State                                     | Current draw |
|-------------------------------------------|--------------|
| Listen / receive (radio RX + MCU active)  | 21.5 mA      |
| Transmit (radio TX at 0 dBm + MCU active) | 19.2 mA      |
| Sleep (radio off, MCU in low-power mode)  | 5.1 $\mu$ A  |

- (a) The radio is never turned off (the node always listens). Compute the lifetime in hours and days. 4 pts

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- (b) With radio duty cycling the node listens for 10 ms every second (1%), transmits one 4 ms packet every 60 s, and sleeps otherwise. Compute the average current and the lifetime in days and years. 8 pts

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- (c) Which state dominates the average current in (b)? What does this say about where energy optimization should focus? 4 pts

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- (d) What does Contiki's **Energest** module measure, and how do you turn its output into energy in joules? 4 pts

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### Question 5 — Medium Access Control

20 points

Four nodes stand on a line: **A — B — C — D**. Each hears only its direct neighbours.

- (a) Using these nodes, give one example of the **hidden terminal** problem and one of the **exposed terminal** problem. Explain what goes wrong in each.

6 pts

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- (b) Receivers wake every  $T_w = 500$  ms. Compare how long a sender keeps its radio busy before the data frame starts under **B-MAC** (long preamble) and **X-MAC** (strobed preamble with early ACK), assuming the receiver's wake-up time is uniformly random. Why does X-MAC also help overhearing nodes?

6 pts

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Six nodes must get TDMA slots. Two nodes may not share a slot if they interfere; the interference edges are:

A-B, A-C, B-C, B-D, C-D, C-E, D-F, E-F

- (c) Find a slot assignment with the minimum number of slots and prove it is minimum.

5 pts

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- (d) In TSCH,  $\text{channel} = \text{seq}[(\text{ASN} + \text{channelOffset}) \bmod N]$ . With hopping sequence  $\text{seq} = [15, 20, 25, 26]$ ,  $\text{ASN} = 1003$  and  $\text{channelOffset} = 2$ , which channel is used? Why does TSCH hop at all?

3 pts

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### Question 6 — 6LoWPAN and RPL Routing

14 points

- (a) Why can't an IPv6 packet be sent directly in an IEEE 802.15.4 frame? Name the two 6LoWPAN mechanisms that solve this.

4 pts

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After header compression a datagram is **400 bytes**. Each 802.15.4 frame leaves **102 bytes** for 6LoWPAN. The first fragment header is 4 bytes, every later one 5 bytes, and every fragment except the last must carry a multiple of 8 bytes of datagram.

(b) How many fragments are sent, and how many datagram bytes does each carry? 5 pts

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In an RPL DODAG, node X has two candidate parents. Rank is based on ETX (sum of link ETX along the path to the root R).

| Candidate parent | ETX of link X → parent | Parent's path ETX to R |
|------------------|------------------------|------------------------|
| A                | 1.2                    | 2.0                    |
| B                | 2.5                    | 1.0                    |

(c) Which parent does X choose, and what is X's path ETX? Would a hop-count metric necessarily choose the same parent? Why is ETX usually the better metric in a WSN? 5 pts

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— END OF EXAM · GOOD LUCK —